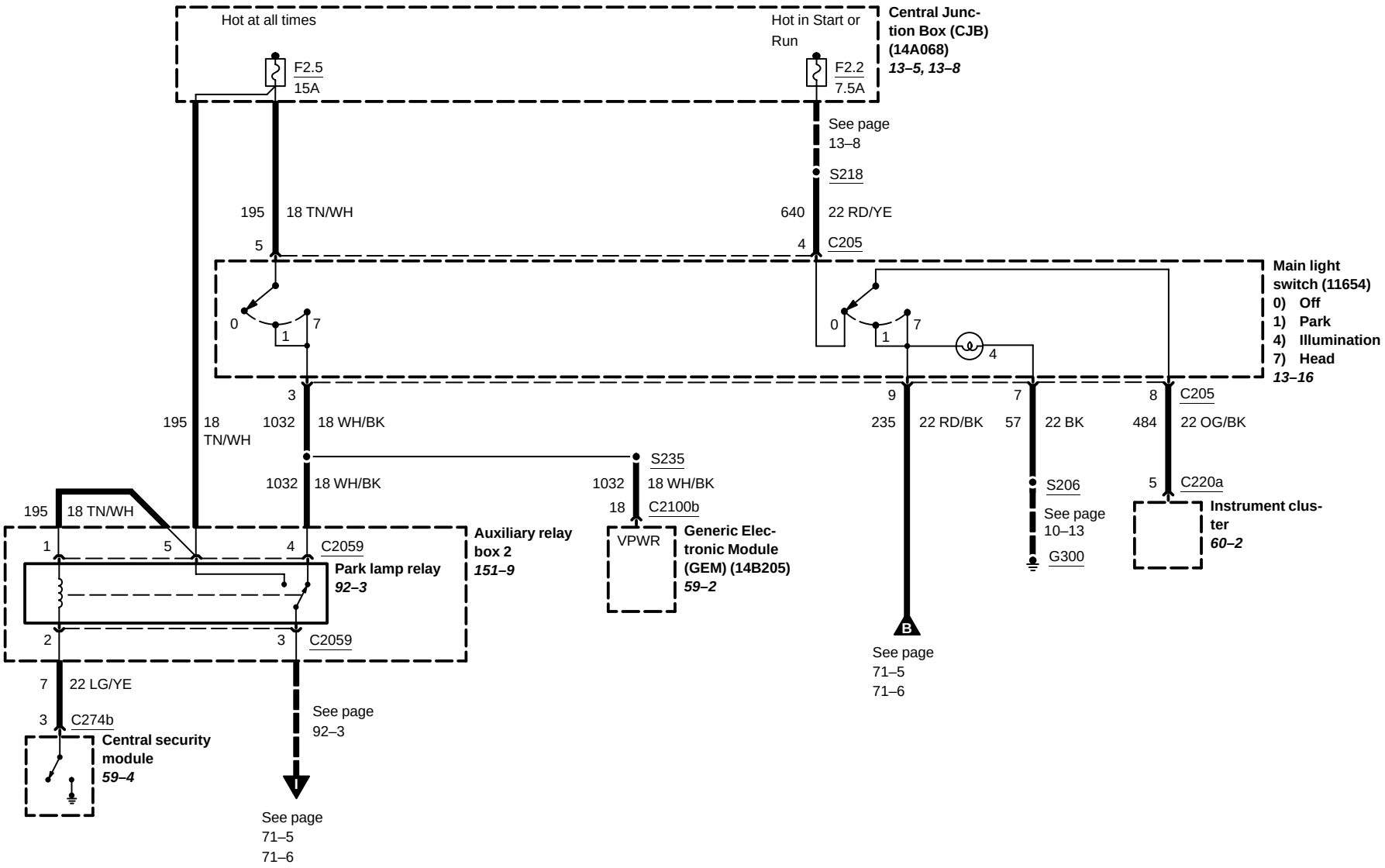without autolamps